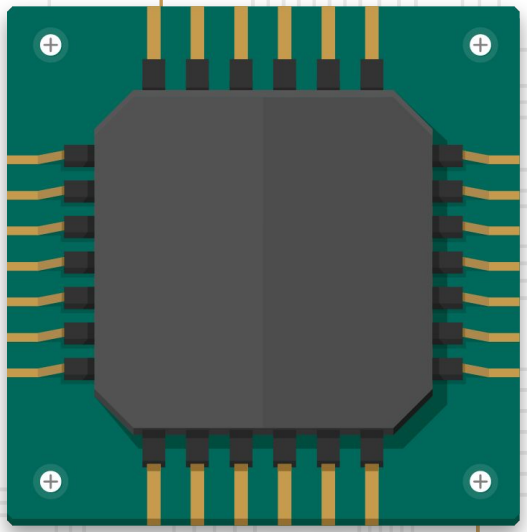
Landslide Barrier Wall

Landon Thoman, Anna Zheng



Project Overview

- **Humanitarian Issue**
  - Region: Nepal’s mountainous communities
  - Unprotected landslide risk





Design Specifications

Requirements:

1) Detect soil moisture & falling debris
2) Deploy barrier if triggered
3) Manual emergency override
4) Include visual warning

Constraints:

1) Remote region
2) Limited materials

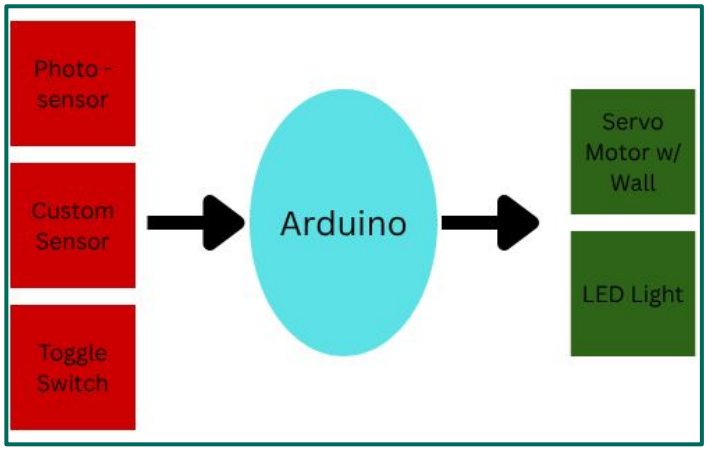
Inputs:

1) Custom Sensor
2) Photoresistor
3) Toggle Switch

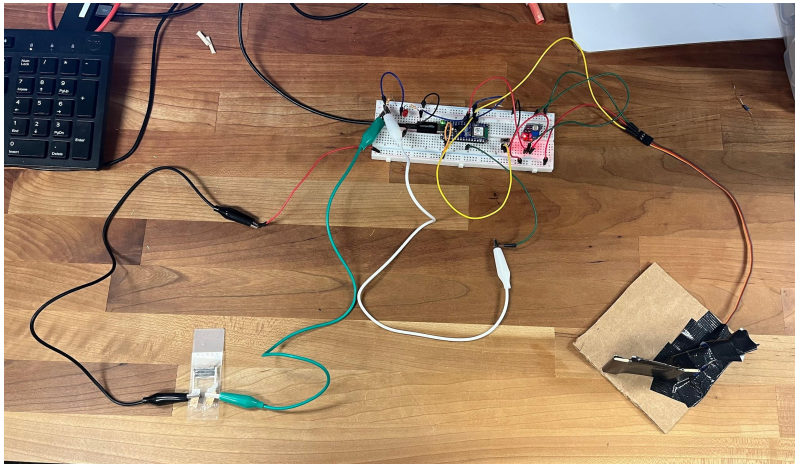
Outputs:

1) Servo Motor
2) LED

Implementation Overview

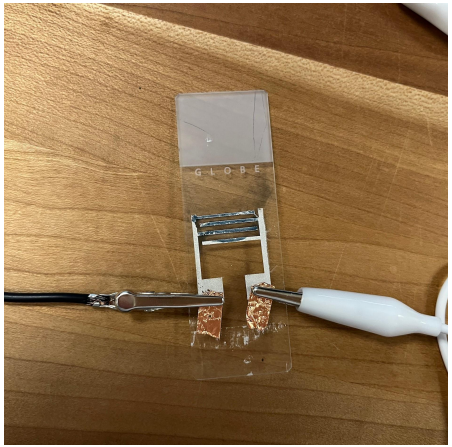


Block Diagram

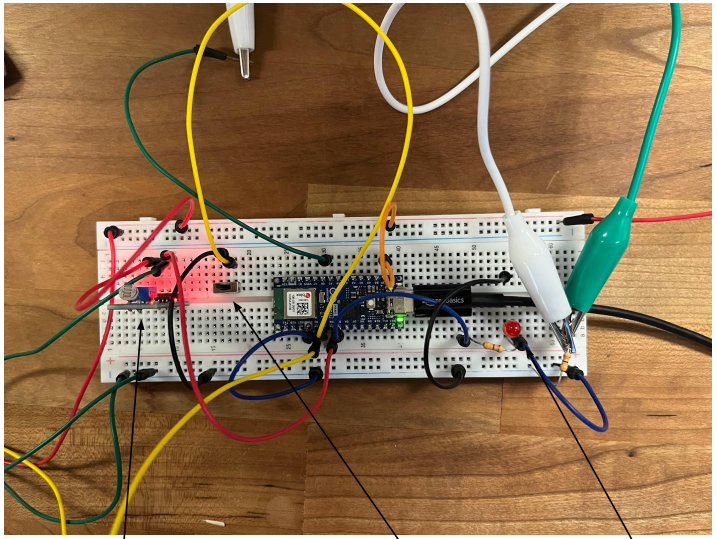


Bird's Eye View of System

Assembly



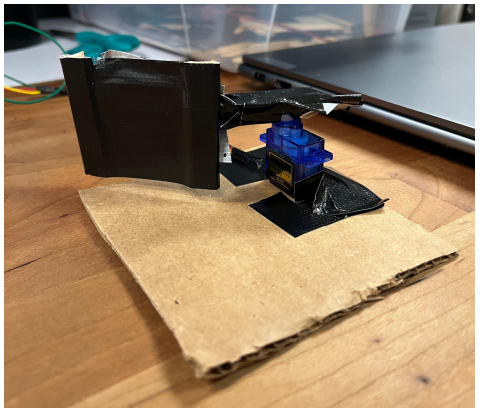
Custom Sensor



Photoresistor

Slide Switch

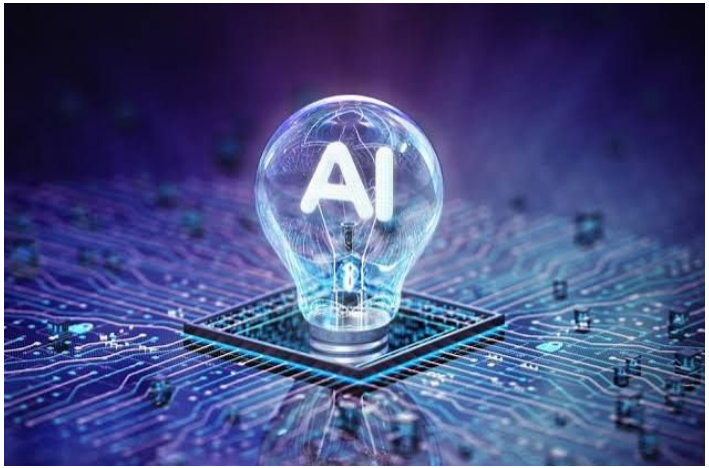
LED



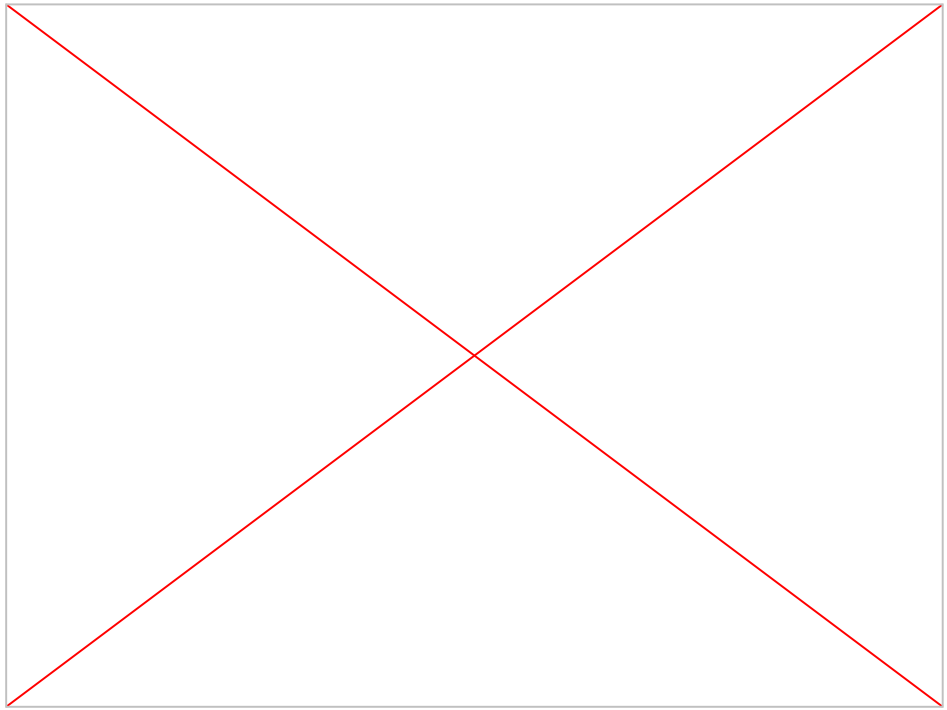
Servo Motor

Use of AI

- **How?**
  - Write MicroPython code
  - Fix circuit wiring
  - Assist wall-to-servo attachment
- **Why?**
  - Increase productivity
  - Catch design errors early
  - Bridge software logic to circuit



Results



1) Detects high moisture
2) < 1 second trigger response
3) 0° - 90° wall deployment
4) Instant LED alert & manual override

Conclusion

Achievement:

- Integrated sensors with working outputs
- Combined automatic deployment with manual override

Next Steps:

- Combine LED warning with audible alarms
- Scale up the sensors & barrier